

# **GLOBAL CO<sub>2</sub> EMISSIONS: TRENDS, REGIONAL SHIFTS, AND PER CAPITA INEQUALITIES**

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1750–2024

# Atmospheric CO<sub>2</sub> Reaches **429 ppm**

**50% Above Pre-Industrial Levels**

# 429 ppm

The February 2026 Mauna Loa measurement marks a >50% increase from the ~280 ppm pre-industrial baseline, an acceleration roughly 1.5x faster than post-ice-age natural warming.

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**Pre-1750**

~280 ppm (Baseline)

**1958**

~315 ppm (Keeling Curve starts)

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**2026**

**429 ppm**

# Global Fossil Fuel CO<sub>2</sub> Emissions Grew **Six-Fold**

From 6 Billion to 35+ Billion Tonnes Since 1950



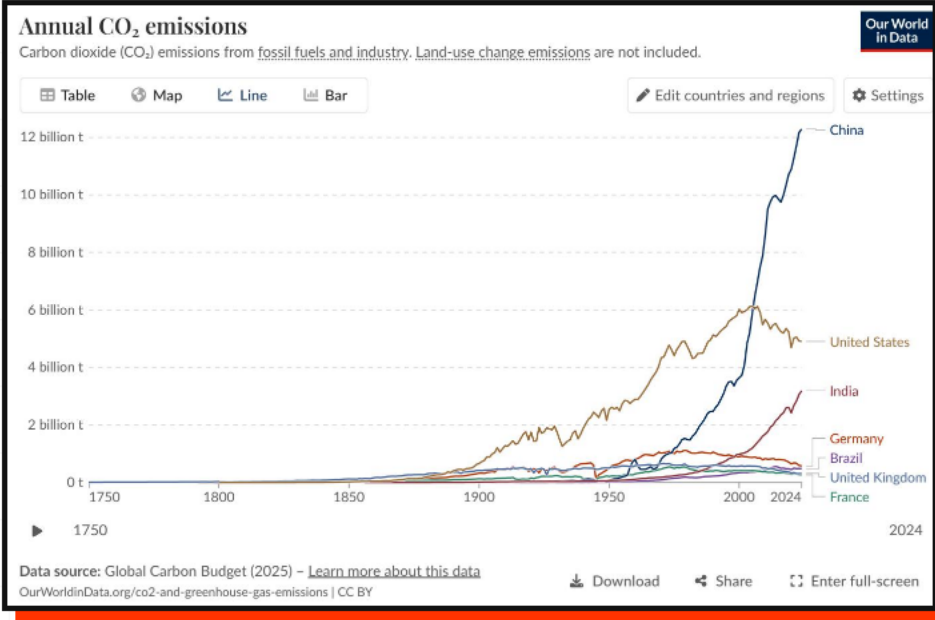
Before the Industrial Revolution, CO<sub>2</sub> emissions were negligible. Today, emissions from fossil fuels and industry exceed 35 billion tonnes. Growth has slowed, but not yet peaked.

# China Now Emits Over 12 Billion Tonnes

More Than Double the United States

| COUNTRY       | ANNUAL CO <sub>2</sub> | TREND           |
|---------------|------------------------|-----------------|
| China         | ~12.5 Bt               | Slight increase |
| United States | ~5.0 Bt                | Declining       |
| India         | ~3.0 Bt                | Rising          |
| EU-27         | ~2.7 Bt                | Declining       |

China accounts for >25% of global emissions. The US peaked near 6 Bt around 2005 and has since declined.



## Asia Accounts for ~50% of Global Emissions

US + Europe Share Falls Below One-Third

1900

>90%

EUROPE + US SHARE

2024

~50%

ASIA'S SHARE

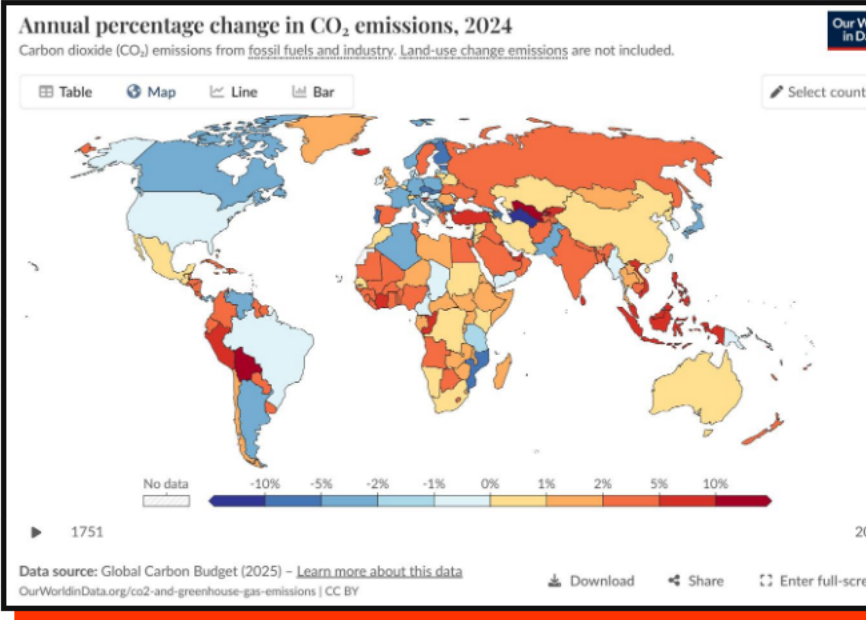
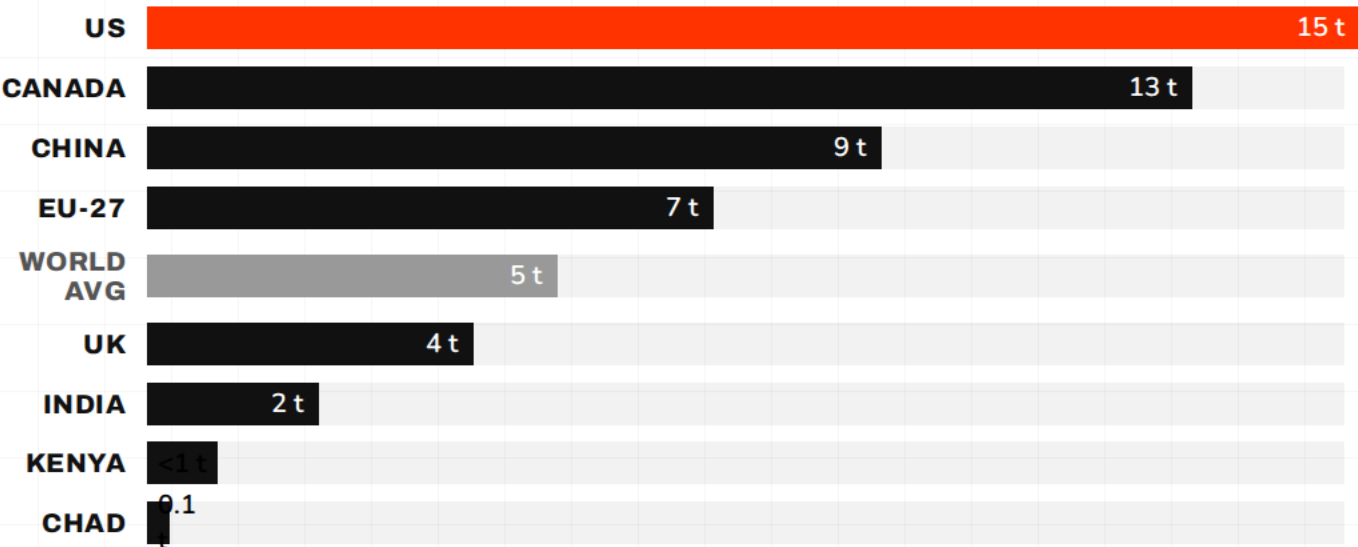
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US + Europe now < 33%

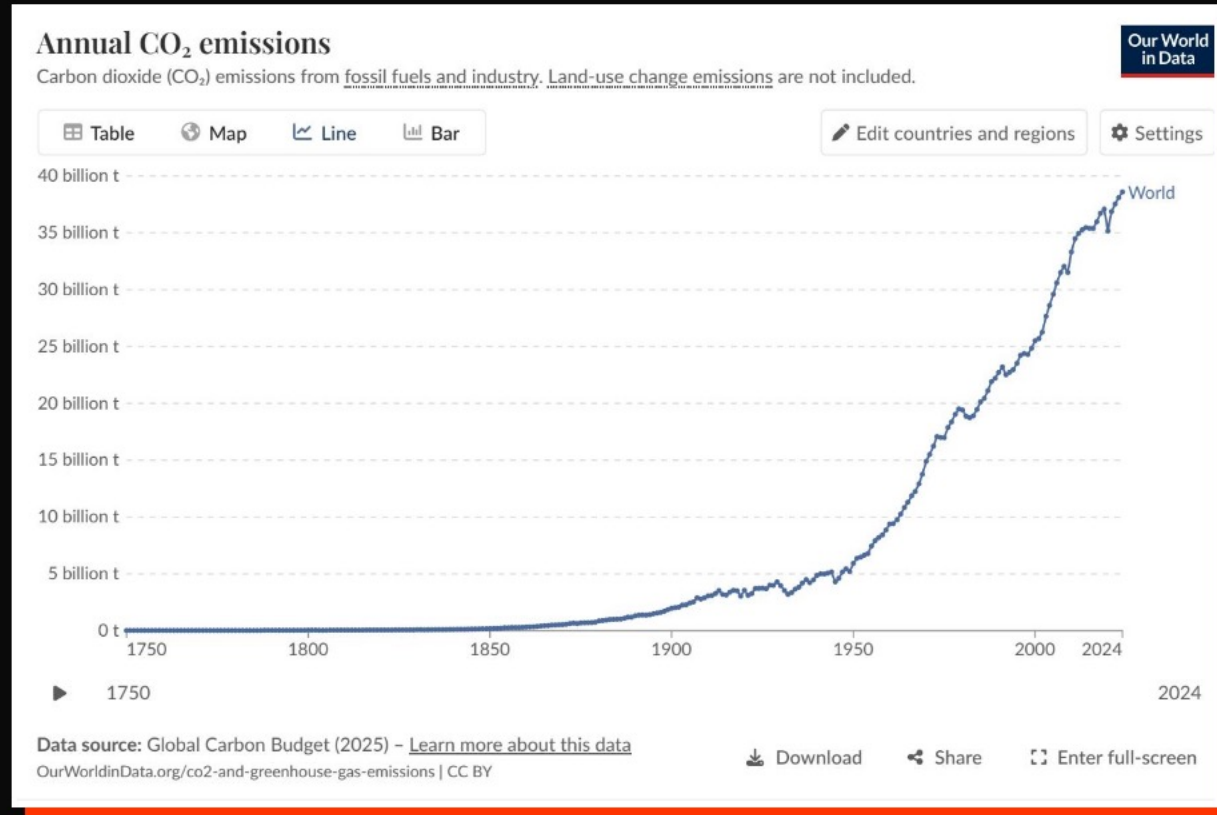
Africa & South America ~ 3-4% each

# Per Capita Gap: 15 t vs. 0.1 t

A 150× Disparity Between the US and Chad



# 2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



**Negative (-1% to -5%)**  
United States and much of Europe

**Positive (+2% to >10%)**  
Russia, Southeast Asia, and several African nations

# Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar prosperity levels can have very different per capita footprints depending on their reliance on nuclear, renewables, or fossil fuels.

## Lower Emissions

| COUNTRY | PER CAPITA | PRIMARY MIX      |
|---------|------------|------------------|
| France  | ~4 t       | High Nuclear     |
| UK      | ~4 t       | Renewables + Gas |

## Higher Emissions

| COUNTRY     | PER CAPITA | PRIMARY MIX         |
|-------------|------------|---------------------|
| Germany     | ~6 t       | Higher Fossil Share |
| Netherlands | >6 t       | Higher Fossil Share |



# Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

Global emissions exceed 35 Bt and have not yet peaked, despite slowing growth rates.

China alone accounts for over 25% of the global total (>12 Bt).

Per capita inequalities remain extreme, with a 150× gap between the highest and lowest emitters.

The US and Europe are declining in annual output, though their historical cumulative responsibility remains large.

Policy and energy mix choices — not just prosperity — determine per capita outcomes.